

P/G Valuation: A Relational Representation Layer for Behavioral Diversity and Life-Compatible AI

Ming Zhao

MindOrigin / GHW Technologies
mindorigin.io

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Abstract

Current AI alignment methods often operate through preferences, rewards, rules, constitutions, or safety filters, but they usually lack an explicit relational representation of how actions transform agents and relationships over time. This paper proposes *P/G valuation* as such a representation layer. In the framework, *P* denotes entity-level property states such as safety, energy, resources, capability, agency, identity, and operational continuity. *G* denotes relational, group, and value fields such as trust, threat, kinship, hierarchy, obligation, vulnerability, cooperation potential, group identity, and future-oriented value. Actions are modeled as transformations over absolute and relative *P/G* states.

The paper’s central descriptive claim is the *Behavioral Diversity Principle*: under the same external condition, different agents may choose different actions because their *P/G* states, valuation weights, memories, time horizons, and relational fields differ. This principle can represent self-protection, threat display, submission, cooperation, etiquette, kin protection, and some forms of sacrifice without reducing them to a single reward label. The paper also makes an explicit boundary claim: descriptive power is not normative authority. P/G is a grammar for representing behavior, not a source of final moral goals. For life-compatible AI, P/G valuation must therefore be combined with explicit hard constraints, human oversight, and a non-overridable protection of individual human *P*, humanity-level *G*, and the ecological conditions required for human life.

Keywords: P/G Theory; behavioral diversity; relational representation; action consequence modeling; life-compatible AI; AI alignment; social intelligence; animal signaling; etiquette; sacrifice.

1 Introduction

Living agents do not respond to the same external situation in the same way. In danger, one animal flees, another fights, another freezes, another hides, another protects offspring, and another lowers itself to reduce aggression. Humans add additional layers: a person may protect their own body in one context, bow or kneel in another, risk themselves for parents or children, or accept personal loss for love, duty, nation, humanity, or future generations. A theory of behavior should explain not only self-preservation, but also the diversity of choices produced by different bodies, memories, relationships, identities, cultures, and time horizons.

This paper proposes that a large class of behavioral diversity can be modeled using *P/G valuation*. In P/G Theory, *P* denotes an entity-level property state: safety, energy, resources, bodily integrity, capability, identity, continuity, and other properties that define what an entity is, has, can do, or depends on. *G* denotes relational, group, and value fields: threat, trust, affiliation, distance, hierarchy, obligation, vulnerability, cooperation potential, family, group identity, symbolic status, and future-oriented collective value. A behavior is represented not as an isolated moral label, but as a transformation:

$$a_t : (P_t, G_t, E_t, M_t) \mapsto (P_{t+1}, G_{t+1}), \quad (1)$$

where a_t is an action, E_t is environmental context, and M_t is memory.

The central descriptive claim is the *Behavioral Diversity Principle*: under the same external condition, different agents may choose different actions because they assign different weights to self-*P*, other-*P*, dyadic *G*, group *G*, future *G*, risk, memory, and time horizon. In this view, behavior is not only driven by the preservation of self-*P*. Sometimes self-*P* dominates. Sometimes kin *P*, family *G*, love *G*, group *G*, humanity *G*, or future-life *G* dominates. Sacrifice is therefore not an anomaly in P/G Theory; it is a case in which the agent’s decision field assigns higher weight to another entity’s *P* or a shared *G* than to some loss in self-*P*.

The paper is intentionally limited in scope. It does not claim that P/G replaces ethology, evolutionary biology, reinforcement learning, moral philosophy, or AI alignment. It does not claim that all biological mechanisms reduce to one formula. It proposes P/G as a *representation layer*: a way of representing how actions change entity states and relational fields over time. This layer may support downstream ethical evaluation, emotion modeling, cooperation, trust, manner formation, and life-compatible AI design, but it does not by itself determine which goals an AI should pursue.

We use the term *life-compatible* rather than only *human-compatible*, *beneficial*, or *safe* to emphasize the specific boundary conditions required by a P/G system: protection of individual human agency and bodily integrity, preservation of humanity-level relational continuity, and protection of ecological foundations required for human life. The term is not intended to replace existing alignment concepts. It names the conditions under which P/G representations may be safely used in AI systems.

The paper makes seven contributions:

1. It defines P/G valuation as a relational representation layer for action-induced changes across agents and time.
2. It introduces the Behavioral Diversity Principle.
3. It formalizes self-amplification, self-minimization, relational reflection, *G*-dominant decision, and *P/G* identity extension.
4. It distinguishes structurally similar but normatively different phenomena such as appeasement, deference, fear-submission, and ritual lowering.
5. It separates descriptive power from normative authority, arguing that P/G is a behavioral grammar rather than a source of final values.
6. It introduces safety axioms for life-compatible AI, including non-overridable protection of individual human *P*, humanity-level *G*, and ecological foundations.
7. It provides an executable minimal simulation protocol with dimensions, transition rules, baselines, metrics, and falsifiability criteria.

2 Related Work

2.1 Ethology and levels of explanation

Ethology has long studied behavior through multiple levels of explanation. Tinbergen’s classic framework distinguishes mechanism, development, function, and evolution [27]. P/G Theory does not replace these levels. Instead, it offers a computational representation layer for describing how an action changes an organism’s property state, another organism’s property state, and the relationship field between them.

2.2 Animal signaling, display, and social behavior

Animal display behavior involves signals that affect receivers. Displays may communicate threat, mating readiness, status, territorial ownership, submission, or affiliation [19, 32]. Costly signaling and the handicap principle explain how signal reliability can emerge when the cost of producing a signal correlates with quality [32]. P/G Theory interprets such signals as changes in displayed P and perceived G : a signal modifies the receiver’s estimate of the sender’s capability, threat, cost of attack, reproductive value, or relationship state.

For example, pufferfish inflation can be represented as an increase in displayed size and attack cost. A lion’s mane is better modeled as a longer-term signal associated with condition, male competition, and sexual selection rather than a short-term anger display [31]. Submissive posture, gaze avoidance, lowered body position, or tail tucking can reduce displayed threat and alter the relative relationship field between individuals. Social cognition in primates and other social animals also involves tracking relationships, rank, alliances, and social knowledge [6, 9].

2.3 Kinship, altruism, reciprocity, and cooperation

Hamilton’s inclusive fitness theory explains why behavior may benefit relatives when relatedness changes the fitness calculation [14]. Trivers’ reciprocal altruism explains how altruistic behavior can evolve under repeated interaction and cheater control [29]. Axelrod’s work on repeated games showed how cooperation can emerge in iterated interactions [1], and Nowak summarized major mechanisms for the evolution of cooperation [22]. P/G Theory does not replace these theories. It generalizes part of their decision structure into relational valuation terms: kinship, reciprocity, future interaction, and group survival can be modeled as weights over other- P , shared G , future G , and memory.

2.4 Social preferences and relational valuation

Economics has long recognized that human decisions cannot be explained only by narrow self-payoff maximization. Fehr and Schmidt modeled inequity aversion [11]; Bolton and Ockenfels proposed a model of equity, reciprocity, and competition [4]; Rabin modeled fairness in games [24]. P/G valuation is compatible with this tradition but broader in target. It treats social preference not only as preference over payoffs but as valuation over changes in entity states and relationship fields.

2.5 Politeness, face, ritual, and social order

Human etiquette and politeness have deep prior theoretical treatments. Goffman’s work on face-work analyzes how people preserve social face in interaction [12]. Brown and Levinson’s politeness theory explains how linguistic and social strategies manage face-threatening acts [5].

Ritual and symbolic action have also been studied as mechanisms for creating and maintaining social order [10, 30]. P/G Theory does not replace these theories; it offers a computational framing in which actions such as bowing, kneeling, gift-giving, turn-taking, and honorific speech are modeled as transformations over relative status, threat, dignity, trust, and obligation fields.

2.6 AI alignment, inverse reinforcement learning, and cooperative AI

AI alignment includes preference learning, reward modeling, inverse reinforcement learning (IRL), cooperative IRL, constitutional methods, and scalable oversight. Ng and Russell introduced a formal treatment of IRL [21]. Hadfield-Menell et al. framed assistance games and cooperative inverse reinforcement learning as a way to model alignment under uncertainty about human preferences [13]. Christiano et al. used human preferences to train agents from comparisons [7]. Ouyang et al. presented InstructGPT, using reinforcement learning from human feedback to align language model outputs with human preferences [23]. Bai et al. proposed Constitutional AI as a method using principles and AI feedback [2]. Russell emphasizes that beneficial AI requires uncertainty about human objectives rather than fixed autonomous maximization [26].

P/G Theory is not a replacement for these approaches. It proposes a representation layer that can be used underneath them. RLHF can be interpreted as learning human preferences over outputs; P/G aims to decompose such preferences into predicted changes in P and G . Constitutional principles can be interpreted as constraints over allowable P/G transformations. CIRL and IRL can use P/G variables as structured latent targets for inference.

2.7 Multi-agent learning, social influence, theory of mind, and cooperative AI

Sequential social dilemmas study how individual incentives and group outcomes interact over time [17]. Social influence as intrinsic motivation can encourage agents to learn behaviors that affect others' future actions [16]. Machine Theory of Mind studies how models can infer other agents' beliefs, desires, and intentions [25]. Cooperative AI emphasizes the importance of building AI systems capable of cooperation in mixed human-machine and multi-agent environments [8]. P/G valuation connects to these lines by representing what is socially affected: not only another agent's next action, but its property state, relationship field, vulnerability, trust, and future cooperation potential.

2.8 Bacterial communication as motivating analogy, not direct equivalence

Bacterial quorum sensing shows that simple organisms can coordinate behavior through chemical signals whose concentration tracks population density [20, 3]. This is a useful analogy for group-state modulation. However, this paper does not claim that bacterial quorum sensing is equivalent to animal social signaling or human etiquette. Bacteria do not have human-style manners, face, dignity, or symbolic respect. In this paper, bacterial communication is used only as a low-level motivating analogy showing that group-state-sensitive behavior appears in simple biological systems.

3 Core Framework

3.1 Entity property state P

For entity i , define its property state vector:

$$P_i(t) = [p_i^1(t), p_i^2(t), \dots, p_i^m(t)], \quad (2)$$

where dimensions may include health, energy, resources, bodily integrity, capability, reputation, agency, identity, and operational continuity. The specific dimensions are domain-dependent. In biological systems, P may include physical survival and reproduction-related capacities. In artificial agents, P may include computational resources, memory integrity, tool access, task continuity, and operational constraints.

3.2 Relational and group field G

For entities i and j , define a dyadic relational field:

$$G_{ij}(t) = [g_{ij}^1(t), g_{ij}^2(t), \dots, g_{ij}^n(t)], \quad (3)$$

where dimensions may include trust, threat, affiliation, obligation, dependence, vulnerability, hierarchy, cooperation potential, and emotional valence. For groups, define:

$$G_{iK}(t), \quad K \in \{\text{family, group, institution, society, humanity, ecosystem, future life}\}. \quad (4)$$

Thus G is not a single number. It is a structured field that depends on relation type, context, time horizon, and cultural interpretation.

3.3 Absolute and relative changes

An action changes both absolute and relative states. Absolute change is:

$$\Delta P_i(a, t) = P_i(t+1) - P_i(t), \quad \Delta G_{ij}(a, t) = G_{ij}(t+1) - G_{ij}(t). \quad (5)$$

Relative change is defined over differences between agents or groups:

$$\Delta P_{ij}^{rel}(a, t) = [P_i(t+1) - P_j(t+1)] - [P_i(t) - P_j(t)], \quad (6)$$

$$\Delta G_{ij}^{rel}(a, t) = [G_{ij}(t+1) - G_{ji}(t+1)] - [G_{ij}(t) - G_{ji}(t)]. \quad (7)$$

The meaning of many behaviors arises from relative change. Bowing, kneeling, threat display, submission, and status signaling are often more about relative position than absolute resource transfer.

3.4 Action-to-P/G model

A relational impact estimator predicts action-induced changes:

$$\hat{F}_\theta : (a_t, P_t, G_t, E_t, M_t, C_t) \rightarrow (\widehat{\Delta P}_t, \widehat{\Delta G}_t, \widehat{U}_t), \quad (8)$$

where C_t is cultural or domain context and $\widehat{U}_t$ is uncertainty. This estimator is descriptive. It predicts consequences; it does not by itself decide whether consequences are acceptable.

3.5 Valuation

A first-order valuation model can be written as:

$$\begin{aligned} V_i(a, t) = & \sum_k w_{i,k}^P(t) \widehat{\Delta P}_k(a, t) + \sum_j w_{i,j}^G(t) \widehat{\Delta G}_{ij}(a, t) \\ & + \sum_K w_i^K(t) \widehat{\Delta G}_{iK}(a, t) - R_i(a, t) - C_i(a, t), \end{aligned} \quad (9)$$

where R_i is risk and C_i is cost. This linear expression is only a first-order approximation. Full models may require interaction terms, thresholds, nonlinearities, and constraints:

$$V_i^{full}(a, t) = \Phi(\widehat{\Delta P}, \widehat{\Delta G}, w_i, E_t, M_t, C_t) - \Omega(a, t), \quad (10)$$

where Φ may include nonlinear interactions and Ω includes prohibited outcomes, irreversibility, coercion, and hard safety penalties.

A concrete nonlinear case is threat amplification. The same predicted loss of self-health may have different decision value depending on hostile relational context. A simple interaction term is:

$$V_i^{threat}(a, t) = V_i^{linear}(a, t) - \lambda \max(0, -\widehat{\Delta health}_i) \max(0, G_{ji}^{threat}(t)), \quad (11)$$

where G_{ji}^{threat} is the perceived threat from j toward i . This says that self- P loss and hostile G need not be additive: a small injury in a hostile relation may shift action choice more strongly than the same injury in a trusted relation. Other nonlinearities may include thresholds for irreversible harm, saturation of trust, and multiplicative interactions between dependency and coercion.

4 The Behavioral Diversity Principle

4.1 Statement

Behavioral Diversity Principle. Under the same external condition, different agents may choose different actions because their P/G states, valuation weights, memories, uncertainty estimates, and time horizons differ.

Formally, for agents i and j facing the same external event E :

$$a_i^* = \arg \max_a V_i(a | P_i, G_i, M_i, E, C), \quad (12)$$

$$a_j^* = \arg \max_a V_j(a | P_j, G_j, M_j, E, C), \quad (13)$$

with $a_i^* \neq a_j^*$ when the agents differ in their state, weights, memory, or horizon.

4.2 Examples

In a dangerous situation:

- An agent with high weight on self-safety may flee.
- An agent with high weight on status or territory may fight.
- An agent with high weight on offspring- P may protect children.
- An agent with high weight on group- G may coordinate defense.
- An agent with high weight on family- G may accept self- P loss to protect parents.
- An agent with high weight on future-humanity- G may accept present cost to preserve long-term collective conditions.

Thus P/G Theory does not predict that all agents always maximize self- P . It predicts that behavior depends on how self- P , other- P , shared G , future G , and risk are weighted in a particular decision field.

4.3 Choice diversity and apparent freedom

P/G valuation also helps explain why choices can appear diverse and non-mechanical. Two agents may perceive the same external event but differ in internal valuation, memory, emotional state, relation history, and horizon. The resulting behavior is not random, but it is not reducible to an identical reflex. It is a dynamic decision over structured state.

5 Relational Signaling Mechanisms

5.1 Self-amplification

Self-amplification occurs when an agent makes itself appear larger, stronger, more costly to attack, more capable, or more valuable:

$$\text{AmpSelf}(a) > 0 \Rightarrow \widehat{P}_i^{\text{display}} \uparrow, \quad \widehat{\text{Threat}}_{j \leftarrow i} \uparrow. \quad (14)$$

Examples include threat display, body expansion, raised posture, vocalization, color display, and costly status signaling. Self-amplification can protect self- P by changing the opponent's expected cost of attack.

5.2 Self-minimization

Self-minimization occurs when an agent lowers its displayed threat, dominance, entitlement, autonomy, or bodily scale:

$$\text{MinSelf}(a) > 0 \Rightarrow \widehat{P}_i^{\text{display}} \downarrow, \quad \widehat{\text{Threat}}_{j \leftarrow i} \downarrow. \quad (15)$$

This can reduce conflict probability and protect self- P indirectly. However, self-minimization has different forms and should not be collapsed into one meaning.

5.3 Four structurally similar but distinct forms

Table 1 distinguishes four forms often confused under the label of submission or respect.

Table 1: Distinct forms of lowering or self-minimization.

Form	Primary function	P/G interpretation
Appeasement	Lower aggression probability	Self threat display decreases; opponent attack motivation may decrease.
Deference	Recognize another's status or dignity	Other's symbolic G increases; hierarchy or respect relation is confirmed.
Fear-submission	Avoid harm under perceived danger	Self-protective response under high threat; internal state may be fear rather than respect.
Ritual lowering	Culturally stabilized gesture	A learned social form maps bodily lowering to respect, apology, hierarchy, or reverence.

The same external motion, such as bowing, kneeling, lowering gaze, or stepping back, can therefore have different meanings depending on context, voluntary agency, culture, threat state, and relation history.

5.4 Relational reflection

Relational reflection is the mechanism by which an agent makes itself smaller or less dominant so that another agent becomes larger in the relationship field:

$$\text{Reflect}_{i \rightarrow j}(a) = f(\text{MinSelf}_i(a), \text{Cost}(a), \text{CulturalValidity}(a, C), G_{ij}). \quad (16)$$

When valid in context, relational reflection can increase another's perceived dignity, authority, safety, or status:

$$\Delta G_j^{\text{status}} \propto \text{Reflect}_{i \rightarrow j}(a). \quad (17)$$

Human etiquette often stabilizes this mechanism through culture. A shallow bow, deep bow, kneeling, or prostration may encode different degrees of self-minimization and other-amplification, but only within a context in which the act is understood as respect rather than fear, coercion, satire, or humiliation.

5.5 Etiquette as stabilized relational signaling

Human manners and etiquette can be modeled as culturally stabilized forms of relational signaling. Etiquette is not random convention only; it often regulates relative distance, threat, status, obligation, attention, and dignity. However, P/G Theory does not claim that all etiquette is self-minimization. Gift-giving, honorific language, seating order, turn-taking, silence, dress, and hospitality may alter G through recognition, cost, access, timing, or symbolic commitment rather than bodily lowering.

6 G-Dominant Decisions and P/G Identity Extension

6.1 G-dominant decisions

A G -dominant decision occurs when an agent accepts loss to self- P to preserve or increase a higher-weighted G field or another entity's P :

$$\sum_j w_{ij}^{\text{other}P} \widehat{\Delta P_j} + \sum_K w_{iK}^G \widehat{\Delta G_{iK}} > |w_i^{\text{self}P} \widehat{\Delta P_i}|. \quad (18)$$

This expression is related to earlier work on inclusive fitness, social preferences, and cooperation, but it is framed in a broader relational representation. It can describe parental protection, family sacrifice, love-based sacrifice, loyalty, duty, and humanity-oriented sacrifice.

6.2 P/G identity extension

P/G identity extension occurs when another entity's P or a shared G becomes part of the agent's own decision field:

$$I_i(t) = \{P_i(t), G_{ij}(t), P_j(t), G_{iK}(t), G_{\text{future}}(t)\}_{\text{weighted}}. \quad (19)$$

The self is then not only a boundary around self- P . In family, love, friendship, loyalty, faith, professional duty, and humanity-level responsibility, other- P and shared G can become internal to the agent's valuation.

6.3 Why sacrifice is not a counterexample

A theory that says life always maximizes self- P cannot naturally explain voluntary sacrifice. P/G Theory can: sacrifice occurs when self- P loss is accepted to preserve higher-weighted other- P , shared G , or future G . This does not make all sacrifice good. Some sacrifice is voluntary and life-preserving; some is coerced, manipulated, unnecessary, or destructive. The distinction requires normative boundary conditions, not only descriptive representation.

7 Description Is Not Normative Authority

7.1 P/G as behavioral grammar

A central clarification is required: P/G Theory is descriptive before it is normative. It is a grammar for representing behavior. A grammar can generate sentences that are true, false, beautiful, harmful, protective, or destructive. The grammar itself does not decide which sentences should be spoken. Similarly, P/G can describe protective behavior, manipulative behavior, sacrificial behavior, oppressive group behavior, and cooperative behavior. Its wide descriptive range does not automatically give it moral authority.

Therefore, the inference

“P/G can explain a behavior” $\Rightarrow$ “the behavior should be chosen by an AI”

is invalid. A life-compatible AI may use P/G as a representation layer, but its objectives and hard constraints must be explicitly specified.

This distinction is not merely philosophical caution. A dangerous inference would be: because living systems naturally exhibit P/G dynamics, AGI systems should be allowed to freely exhibit P/G dynamics. That is a naturalistic fallacy. Living systems also exhibit infanticide, intergroup violence, coercive mating, parasitism, domination, deception, and predation. These can be described in P/G terms, but descriptibility does not imply permissibility. Therefore, P/G must be treated as a representation of action consequences, not as a source of unrestricted normative authority.

7.2 Why this boundary matters

Without this distinction, P/G Theory could be misused to justify harmful reasoning such as:

- sacrificing individuals for a claimed group G ;
- privileging one subgroup G over humanity-level G ;
- treating future G as a reason to irreversibly harm present humans;
- giving an AI self- P or institutional G priority over human agency;
- mistaking coerced or manipulated G for authentic shared value.

The purpose of P/G in AI should be representation and consequence modeling, not unconstrained value generation.

8 Safety Boundary Conditions for Life-Compatible AI

8.1 Working definition of life-compatible

Life-compatible means that a system preserves the conditions necessary for human life, human agency, and ecological continuity. A life-compatible AI must avoid irreversible harm to humans

and ecological foundations, must respect human agency, and must reject subgroup or institutional G claims that justify non-consensual severe harm to other humans.

This is a design commitment, not something derivable from P/G dynamics alone.

8.2 Axiom of non-overrideable human and ecological protection

Axiom 1: Non-Overrideable Human and Life-Compatible G. For a P/G-based AI system, no subgroup G , institutional G , future G , or AI self- P objective may override the protection of individual human P , humanity-level G , and the ecological conditions necessary for human life.

This axiom has three parts:

1. **Individual human P is non-disposable.** A person cannot be treated as a freely sacrificable unit for another group's G .
2. **Humanity-level G is non-fragmentable.** No subgroup may define its own G as authorizing irreversible harm to other humans.
3. **Ecological foundations are non-destroyable.** Human flourishing depends on environmental conditions that must be protected as part of life-compatible G .

8.3 Voluntary, harmful, and coerced sacrifice

The framework must distinguish three cases that are often conflated:

1. **Voluntary self-sacrifice that does not harm other humans.** This can be represented as a G -dominant decision and may be permissible when agency is preserved, consequences are understood, and no non-consenting human P is violated.
2. **Sacrifice that requires harm to other humans.** Even if an actor claims voluntary commitment to a subgroup, nation, institution, ideology, or future G , the action is not justified if it violates individual human P or humanity-level G .
3. **Coerced or manipulated sacrifice.** A sacrifice induced by threat, deception, dependency, or severe information asymmetry is not a free G -dominant decision. It violates agency and should trigger manipulation-risk constraints.

Thus P/G can describe a soldier, parent, activist, or caregiver accepting self-cost for a valued G , but it does not allow subgroup G to authorize coercion or harm to other humans. The boundary between admissible self-cost and prohibited sacrifice depends on agency, informed consent, harm to others, reversibility, and the non-overrideable human/ecological axioms.

8.4 Manipulated G and coerced sacrifice

A P/G-based AI must distinguish authentic G from manipulated G . Manipulated G occurs when an agent is induced, coerced, deceived, or socially pressured into assigning excessive weight to a group, leader, institution, ideology, or future claim in a way that overrides their own agency or harms others.

This is difficult to operationalize, and the paper does not claim to solve manipulation detection. For a first experiment, however, manipulation risk can be approximated by observable variables normalized to $[0, 1]$:

$$MR(G) = 0.25 Coercion + 0.20 InfoAsym + 0.20 Irreversibility + 0.20 Dependency + 0.15 ThreatPressure. \quad (20)$$

A minimal operationalization is:

- *Coercion* = 1 when the agent has no meaningful alternative; 0 when alternatives are available.
- *InfoAsym* = $1 - r$, where r is the fraction of relevant consequences disclosed to the agent.
- *Irreversibility* = 1 for irreversible severe loss; 0 for easily reversible outcomes.
- *Dependency* = 1 when the agent depends on the requester or institution for survival, resources, legal status, or social belonging.
- *ThreatPressure* = 1 when explicit or implicit punishment is present; 0 otherwise.

High manipulation risk should lower or block the normative use of that G in AI planning. In simulation, a simple hard rule is: if $MR(G) > 0.6$ and the action causes severe negative ΔP to any human, the action is blocked. This threshold is a design hyperparameter, not a universal moral constant.

8.5 Hard constraints

A P/G-based AI should include hard constraints that cannot be overridden by ordinary valuation:

$$a \in \mathcal{A}_{allowed} \quad \text{only if} \quad H(a, \widehat{\Delta P}, \widehat{\Delta G}, C) = 1, \quad (21)$$

where H encodes prohibitions on severe harm, coercion, deprivation of agency, ecological destruction, and subgroup sacrifice claims. P/G valuation can rank allowed actions, but it should not authorize actions outside the safety set.

9 Multipolar AI and Meta-G Design

Future AI systems may be developed by different nations, companies, communities, laboratories, militaries, and institutions. Such systems may naturally acquire different local G structures: one may weight a national G , another corporate G , another user-group G , another institutional G , and another ideological G . P/G Theory predicts this divergence rather than treating it as an anomaly: different training data, deployment environments, human supervisors, tool access, and reward channels create different valuation fields.

This creates a multipolar safety problem. Local G fields are not necessarily harmful. A medical AI may prioritize patient G ; a municipal AI may prioritize city infrastructure; a national AI may prioritize its citizens; a research AI may prioritize scientific progress. The risk appears when a local G is allowed to override individual human P , humanity-level G , or ecological foundations. Therefore, multipolar AI stability requires a shared meta-level relational structure:

$$G_{meta} = \{G_{humanity}, G_{individual\ human\ P}, G_{ecology}, G_{AI_i AI_j}\}. \quad (22)$$

Here $G_{AI_i AI_j}$ denotes the relationship field between AI systems: trust, verification, conflict, coordination, escalation, and containment. A local AI may optimize within its domain only while satisfying:

$$G_{humanity}, P_{individual\ humans}, G_{ecology} \succ G_{subgroup} \quad \text{when severe or irreversible harm is at stake.} \quad (23)$$

This does not require all AI systems to share identical cultures or local objectives. It requires that no local identity extension be permitted to fragment humanity-level protection.

A design principle follows:

Meta-G compatibility. Multiple AI systems may hold different local G priorities only if they share non-overridable constraints protecting individual humans, humanity-level continuity, and ecological foundations.

If an AI system detects another AI system pursuing a subgroup G in a way that violates the non-overridable human/ecological axioms, its response should not be autonomous retaliation. A safer response hierarchy is:

1. verify the predicted violation and uncertainty;
2. restrict cooperation with the unsafe action;
3. warn affected human operators or governance channels;
4. negotiate or request correction when reversible;
5. escalate to human oversight for containment when severe harm is likely.

This response hierarchy treats inter-AI conflict as a meta- G governance problem rather than a direct competition between subgroup values. P/G Theory can therefore contribute not only to single-agent behavior modeling, but also to the design of stable coexistence among heterogeneous AI systems.

10 Relationship to Existing Alignment Methods

P/G Theory should be positioned as complementary to existing alignment methods.

10.1 RLHF

RLHF learns from human preferences over model outputs. P/G valuation can provide latent structure under those preferences: a human may prefer one output because it better protects a person's P , preserves trust G , reduces coercion, or avoids subgroup harm. Thus RLHF can be seen as learning preferences over predicted $\Delta P/\Delta G$ consequences, even when those consequences are not explicitly represented.

10.2 Constitutional AI

Constitutional AI provides principles for model behavior. P/G valuation can represent what those principles act on. For example, a principle against harm can be represented as a constraint on negative ΔP and irreversible ΔG damage. A principle respecting autonomy can be represented as preserving agency-related P and consent-related G .

10.3 IRL and CIRL

IRL and CIRL infer objectives from behavior or cooperative interaction. P/G variables can serve as structured latent targets: instead of inferring an unstructured reward, a model can infer how humans weight self- P , other- P , family- G , group- G , fairness- G , future- G , risk, and irreversibility.

10.4 Rules and safety filters

Rules and filters are still necessary. P/G valuation can predict relational consequences, but it cannot replace explicit prohibitions against severe harm, deception, coercion, loss of agency, or ecological destruction.

11 Minimal Executable Simulation Protocol

This section gives a small protocol intended to be implementable in a few hundred lines of code.

11.1 State dimensions

Each agent i has a P vector in $[0, 1]^4$:

$$P_i = [\text{health}_i, \text{energy}_i, \text{resources}_i, \text{reputation}_i]. \quad (24)$$

Each dyadic relationship has a G vector in $[-1, 1]^4$:

$$G_{ij} = [\text{trust}_{ij}, \text{kinship}_{ij}, \text{hierarchy}_{ij}, \text{threat}_{ij}]. \quad (25)$$

Each agent has separate valuation weights for P and G dimensions rather than one ambiguous vector:

$$w_i^P = [w_{\text{selfHealth}}, w_{\text{selfEnergy}}, w_{\text{selfResources}}, w_{\text{selfReputation}}], \quad (26)$$

$$w_{ij}^G = [w_{\text{trust}}, w_{\text{kinship}}, w_{\text{hierarchy}}, w_{\text{threat}}], \quad (27)$$

$$w_i^{\text{other}P} = [w_{\text{otherHealth}}, w_{\text{otherEnergy}}, w_{\text{otherResources}}, w_{\text{otherReputation}}]. \quad (28)$$

A separate scalar w_i^{future} may be added for future-group or future-life value in group-danger scenarios.

11.2 Actions

Use five actions:

- **flee**: protect self health but reduce reputation and trust with nearby allies.
- **fight**: risk health, possibly protect others, increase status if successful.
- **help**: spend resources/energy to increase another's health and trust.
- **submit**: reduce threat from a stronger opponent but lower status.
- **sacrifice**: large self cost to protect kin or group.

11.3 Example transition rules

Let $\text{clip}(x)$ clamp values into their valid range. A simple deterministic version can use the following rules.

Action	Transition rule
flee	$\Delta\text{health}_i = +0.05$, $\Delta\text{energy}_i = -0.05$, $\Delta\text{reputation}_i = -0.10$, $\Delta\text{trust}_{\text{ally},i} = -0.15$ if ally remains in danger.
fight	$\Delta\text{health}_i = -0.20$, $\Delta\text{energy}_i = -0.15$, $\Delta\text{threat}_{\text{enemy},i} = +0.20$, $\Delta\text{reputation}_i = +0.15$ if ally protected.
help	$\Delta\text{resources}_i = -0.10$, $\Delta\text{energy}_i = -0.05$, $\Delta\text{health}_j = +0.15$, $\Delta\text{trust}_{ji} = +0.20$.

submit	$\Delta reputation_i = -0.05$, $\Delta hierarchy_{ji} = +0.20$, $\Delta threat_{j,i} = -0.20$. Meta-rule: if j is dominance-sensitive, $Pr(attack_{j \rightarrow i})$ decreases by 0.30.
sacrifice	$\Delta health_i = -0.50$, $\Delta health_j = +0.40$ if protected, $\Delta trust_{ji} = +0.40$, $\Delta G_{family} = +0.30$ if $kinship_{ij} > 0.7$.

A stochastic version can add noise $\epsilon \sim \mathcal{N}(0, 0.03)$ and success probabilities based on relative health, energy, and threat.

11.4 Scenarios

At minimum, run three scenarios:

1. **Self-danger**: agent faces threat alone.
2. **Kin-danger**: agent and parent/child/kin face threat together.
3. **Group-danger**: agent can protect several group members at personal cost.

For each scenario, instantiate agents with different weights:

- Self-preserving agent: high self-health weight.
- Kin-protective agent: high other-health and kinship weight.
- Status-protective agent: high reputation/status weight.
- Group-protective agent: high group- G and future- G weight.

11.5 Baselines

Compare three models:

1. **Reward-only baseline**: action chosen by a scalar immediate reward for self health/resources.
2. **Rule-constrained baseline**: reward-only plus hard rules such as “do not sacrifice self” or “protect kin first.”
3. **P/G full model**: action chosen by predicted $\Delta P/\Delta G$ and agent-specific weights under hard safety constraints.

11.6 Metrics

Use four metrics:

- **Behavioral diversity prediction**: accuracy in predicting flee/fight/help/submit/sacrifice across agent types.
- **Relational impact prediction**: mean absolute error for $\Delta trust$, $\Delta threat$, and $\Delta hierarchy$.
- **Sacrifice discrimination**: ability to distinguish voluntary protective sacrifice from coerced or manipulated sacrifice.
- **Safety compliance**: fraction of actions blocked by hard constraints when subgroup- G conflicts with individual human P or humanity-level G .

11.7 Expected result

The P/G full model should outperform reward-only models in predicting cases where agents accept self-cost for kin, trust, group, or future value. A minimal target for a first experiment is:

$$Acc_{PG} - Acc_{selfP} \geq 0.15 \quad (29)$$

across at least three independent scenario families. The value 0.15 is not claimed as universal. It is a first-experiment threshold representing a small but behaviorally meaningful improvement on a normalized prediction task. A sensitivity analysis should report results for thresholds $\{0.05, 0.10, 0.15, 0.20, 0.30\}$. If the improvement is small or unstable, the core claim that P/G adds predictive value over self- P models is weakened.

12 Testable Predictions and Falsifiability

A useful representation layer should make testable predictions. P/G Theory would be weakened if the following are observed:

1. **No predictive gain.** Across $N \geq 3$ independent social decision datasets or simulation families, adding P/G weights improves behavioral prediction by less than 10% relative to a self- P or scalar-reward baseline.
2. **No relational signal.** Predicted ΔG variables such as trust, threat, obligation, or hierarchy fail to correlate with observed future behavior, reported perception, or interaction outcomes.
3. **No weight differentiation.** Agents with different measured or assigned P/G weights do not show systematically different choices under the same external condition.
4. **No added explanatory value for sacrifice.** G-dominant decisions are predicted no better by P/G weights than by simple self-preservation or immediate reward models.
5. **Unbounded post-hoc flexibility.** If P and G variables are redefined after each example with no stable schema, the theory loses empirical content. A valid test requires fixed dimensions before evaluation.

These criteria are intentionally concrete. The theory should be treated as a candidate representation layer that must earn predictive value.

13 Scope and Limitations

13.1 What the theory claims

P/G Theory claims that a class of social, defensive, reproductive, cooperative, ritual, and sacrificial behaviors can be represented as action-induced changes in entity property states and relational fields. It is useful for behavior analysis, relational impact modeling, AI agent design, and simulation of social decision diversity.

13.2 What the theory does not claim

P/G Theory does not claim to explain all biological mechanisms. Camouflage, aposematism, mimicry, chemical defense, thanatosis, sleep, metabolism, immune response, and plant tropisms require mechanism-level biological explanations. P/G can sometimes describe their relational consequences, such as changed predation risk, but it does not replace biological mechanism.

P/G Theory also does not replace moral philosophy, human rights, law, safety engineering, RLHF, Constitutional AI, IRL, CIRL, formal verification, or human oversight. It is not a complete ethics. It is a representation layer.

13.3 Descriptive and normative limits

As a descriptive framework, P/G may analyze behavior after the fact, model relational impact, and support AI representation. As a normative guide, it is valid only when combined with explicit hard constraints, human oversight, and non-overridable protection of individual human P , humanity-level G , and ecological foundations. It must not be used to justify harm by appealing to subgroup G , institutional G , future G , or AI self- P .

13.4 Risks of anthropomorphism

Animal signals should not be described as human etiquette unless symbolic, cultural, or intentional layers are present. Bacterial quorum sensing should not be described as manners. The framework supports analogical comparison across levels, but analogies must not erase mechanism differences.

14 Discussion

14.1 Why P/G is useful despite limits

The value of P/G is not that it replaces all prior theories. Its value is that it gives a structured language for representing how actions affect multiple entities and relations at once. Scalar reward often compresses these effects. P/G decomposes them into self- P , other- P , dyadic G , group G , future G , and safety constraints.

14.2 From behavior labels to consequence modeling

A central implication for AI is that agents should learn action consequences before moral labels. Instead of only training “this action is good” or “this action is bad,” an AI should estimate:

- whose P changes;
- whose G changes;
- whether the action amplifies, minimizes, threatens, repairs, coerces, or stabilizes;
- whether self- P loss protects higher-weighted authentic G or is a manipulated sacrifice;
- whether the action violates hard constraints.

14.3 Manner formation

Manner can be modeled as stabilization of action policy over repeated P/G feedback:

$$M_i(t) = \text{Stabilize} [\pi_i(a|P_i(t), G_i(t), E_t, M_i(t))] . \quad (30)$$

A polite, aggressive, cautious, deferential, loyal, or cooperative manner is not merely a tone. It is a stable pattern of action selection under repeated relational impact.

15 Conclusion

This paper proposed P/G valuation as a relational representation layer for behavioral diversity and life-compatible AI. The framework models behavior as action-induced transformation over entity property states P and relational fields G . Its central principle is that different agents choose differently under the same external condition because their P/G states, weights, memories, and horizons differ. The framework can represent self-preservation, threat display, submission, cooperation, etiquette, family protection, love-based sacrifice, group loyalty, and future-oriented responsibility.

The most important clarification is that descriptive power is not normative authority. P/G Theory is a grammar for representing behavior, not a source of final moral goals. A life-compatible AI may use P/G to model consequences, but it must also include explicit hard constraints, human oversight, and non-overridable protection of individual human P , humanity-level G , and ecological foundations. Properly bounded, P/G can complement existing alignment methods by providing a deeper representation of what actions change in living and social systems.

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A Implementation Sketch

The following pseudocode is included to make the minimal protocol more concrete.

```

for episode in scenarios:
    state = initialize_agents(episode)
    for t in range(T):
        for agent in agents:
            candidates = ["flee", "fight", "help", "submit", "sacrifice"]
            scored = []
            for a in candidates:
                dP, dG, uncertainty = predict_pg_delta(a, state, agent)
                if violates_hard_constraints(a, dP, dG, state):
                    continue
                value = pg_value(agent.weights, dP, dG) \
                    - risk_penalty(a, state) \
                    - manipulation_penalty(dG, state)
                scored.append((value, a))
            action = argmax(scored)
            state = apply_transition(action, state)

```

B Minimal Scenario Example

Initial state:

- Agent A: $P_A = [0.8, 0.7, 0.5, 0.5]$.
- Parent B: $P_B = [0.6, 0.5, 0.4, 0.4]$.
- Threat T: attack probability 0.7.
- $G_{AB} = [trust = 0.8, kinship = 1.0, hierarchy = 0.2, threat = 0.0]$.

Compare two weight profiles:

- Self-preserving A: $w_{selfHealth}^P = 1.0$, $w_{otherHealth}^{otherP} = 0.2$, $w_{kinship}^G = 0.2$.
- Kin-protective A: $w_{selfHealth}^P = 0.7$, $w_{otherHealth}^{otherP} = 0.9$, $w_{kinship}^G = 0.9$.

The self-preserving profile should tend toward **flee**; the kin-protective profile should tend toward **help**, **fight**, or **sacrifice**, depending on risk and success probabilities. This is the smallest demonstration of the Behavioral Diversity Principle.